RESEARCH ON RESEARCH INSTITUTE

RoRI Consortium Meeting INFLECTION workshop

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Tom Stafford
t.stafford@researchonresearch.org
tomstafford.github.io

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Defining Agentic AI
Effects of Agentic AI
Our response

Artificial Intelligence in four moves

1956 **Calculating machines**

1986 **Learning algorithms**

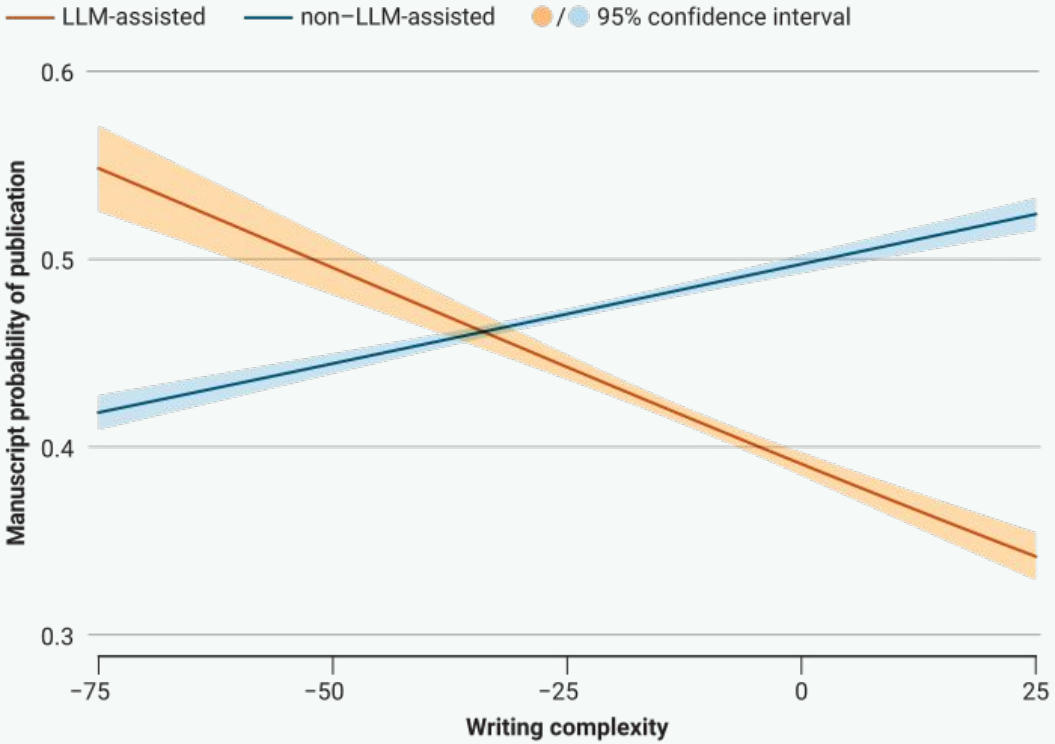
2022 **Large Language models**

2025 **Agents**

i. Agency
ii. Context
iii. Ownership

Scientific production in the era of large language models

https://doi.org/10.1126/science.adw3000



Signal loss

+ Increased productivity
+ Gain greatest for non-native english speakers
+ More diverse literature cited

Diversity Loss https://doi.org/10.1038/s41562-026-02550-0

nature human behaviour

Article

https://doi.org/10.1038/s41562-026-02550-0

The shrinking landscape of linguistic diversity in the age of large language models

Received: 17 February 2025

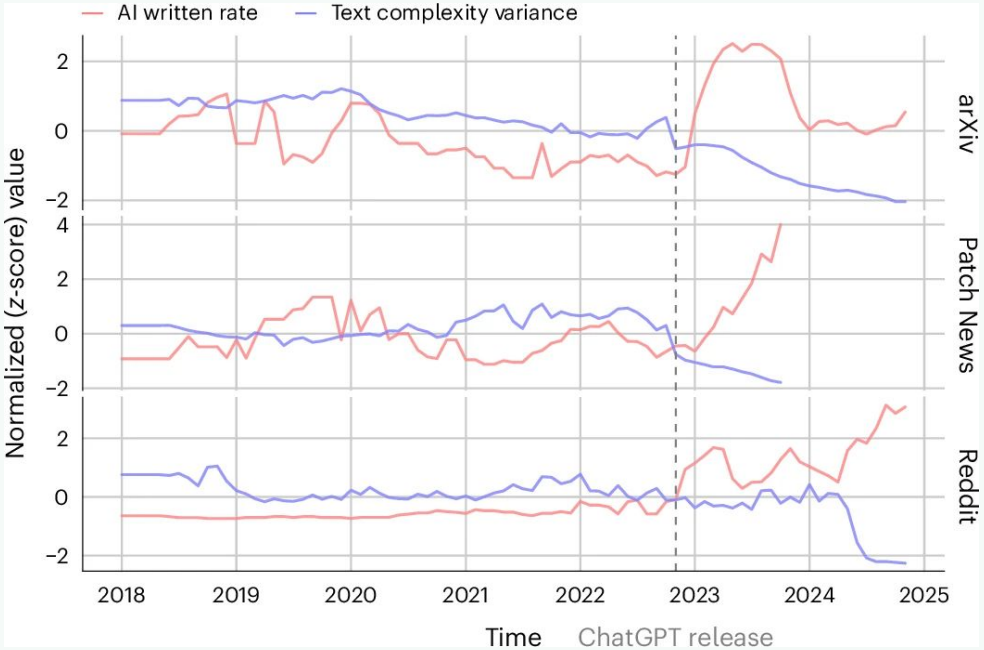
Accepted: 10 July 2026

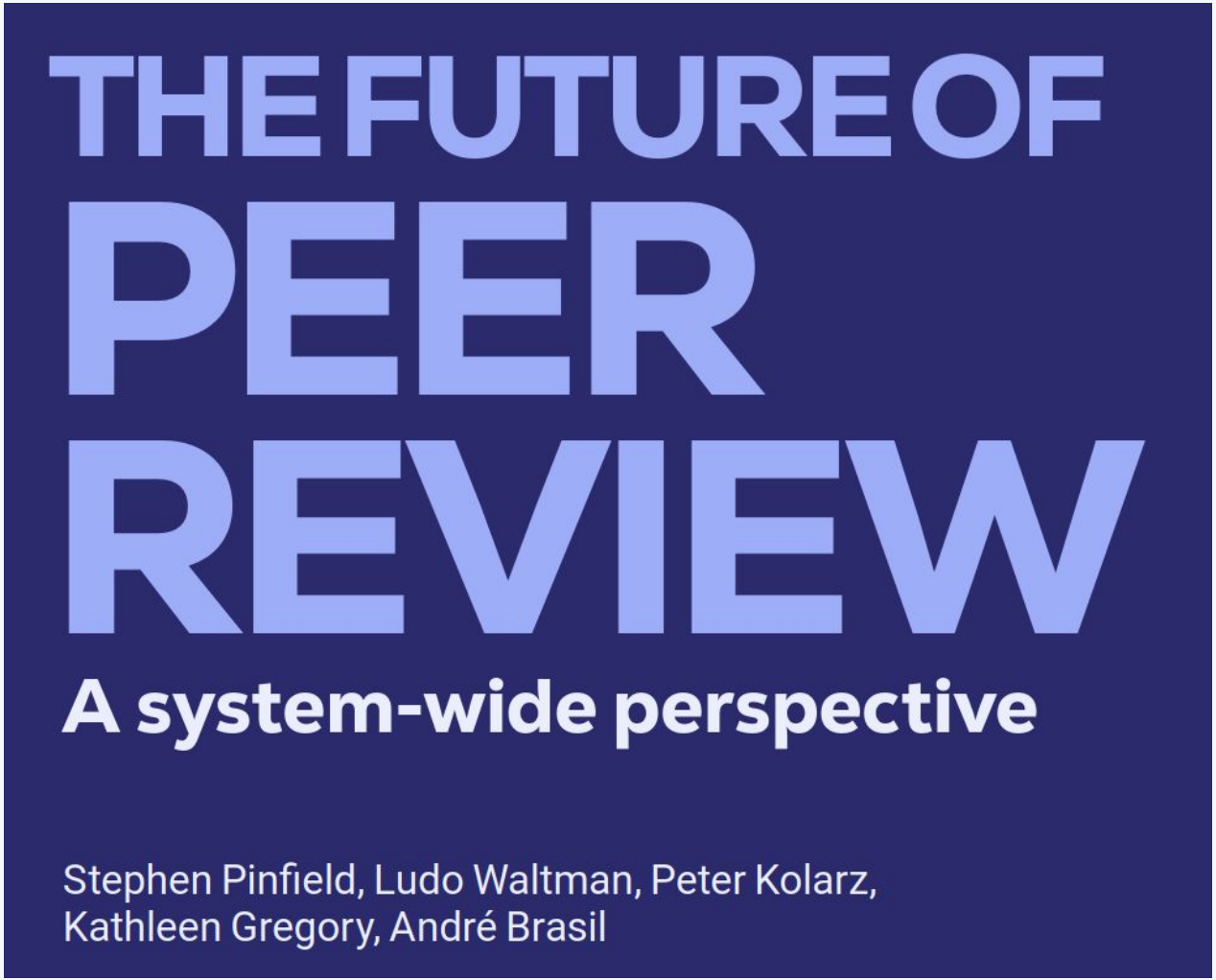
Published online: 24 August 2026

Check for updates

Zhivar Sourati[1,2], Farzan Karimi-Malekabadi[2,3,5], Meltem Ozcan[3,5], Colin McDaniel[3], Alireza Ziabari[1,2], Jackson Trager[2,3], Ala N. Tak[1], Meng Chen[3], Fred Morstatter[1,4] & Morteza Dehghani[1,2,3]

Language is far more than a communication tool; it encodes a wealth of information about a person's identity, psychological state and social context, providing valuable insights for diverse fields including psychology, marketing and healthcare. Across three studies spanning seven datasets in different domains and over 880,000 texts, we show that the widespread adoption of large language models (LLMs) as writing assistants is linked to declines in linguistic diversity, interfering with the societal and psychological insights language provides. While core content is retained when LLMs polish and rewrite texts, LLMs also homogenize writing styles, reducing writing-complexity variance by a statistically significant 21–50% across datasets and models ($P \leq 0.05$), and amplify patterns associated with dominant characteristics while suppressing others, emphasizing conformity over individuality. These trends hold across different LLMs, prompts and contexts, with potential implications for diagnostic processes, personalization efforts, hiring assessments and cultural preservation.





The Future of Peer Review:
A system-wide perspective
https://doi.org/10.6084/m9.figshare.32685351

WP1	WP2	WP3
Analyse	Experiment	Respond

WP1	WP2	WP3
Analyse	**Experiment**	**Respond**
Applicants	Researchers	Policy

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Thank you

Tom Stafford
t.stafford@researchonresearch.org

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RESERVE SLIDES FOLLOW

(not part of talk)

Artificial Intelligence in four moves

1956	**Calculating machines**	**….. Logic engines**
1986	**Learning algorithms**	**….. Neural networks**
2022	**Large Language models**	**….. Chat bots**
2025	**Agents**	**….. “Bureaucratic natives”**

i. Agency
ii. Context
iii. Ownership

Absorptive Capacity

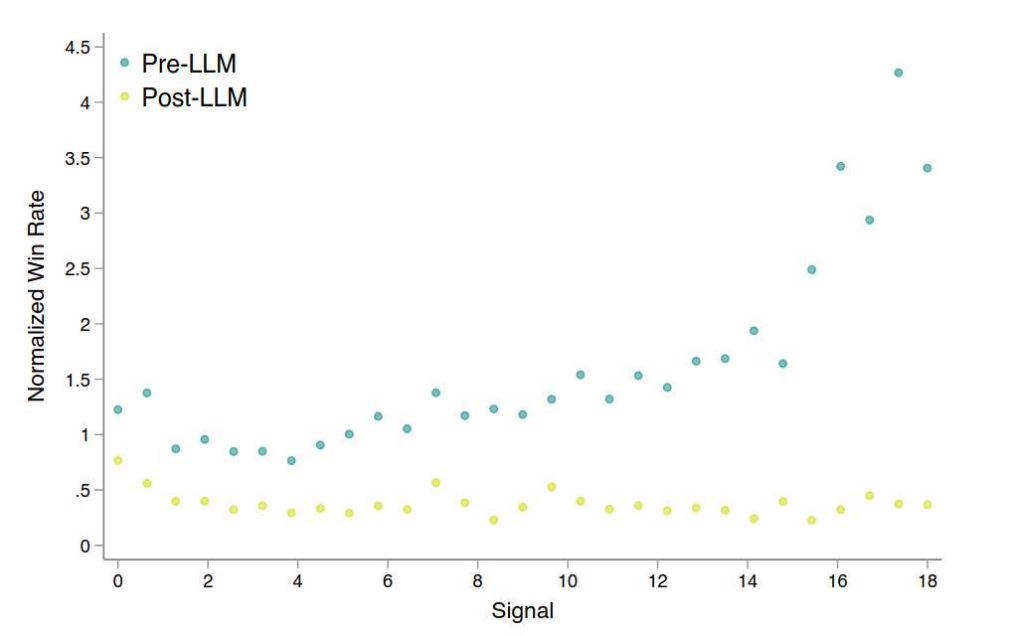
"the ability of a firm to recognize the value of new, external information, assimilate it, and apply it to commercial ends is critical to its innovative capabilities. We label this capability a firm's absorptive capacity and suggest that it is largely a function of the firm's level of prior related knowledge"

Cohen, W. M., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. Administrative Science Quarterly, 35(1), 128–152.

"That is, the performance of basic research may be thought of as a ticket of admission to an information network."

Rosenberg, N. (1990). Why do firms do basic research (with their own money)? Research Policy, 19(2), 165–174.

Making Talk Cheap: Generative AI and Labor Market Signaling. Galdin & Silbert,(2025).arXiv preprint arXiv:2511.08785.



https://tomstafford.substack.com/p/when-trusted-signals-collapse

WP2 an ‘agentic funding sandpit’

“an extreme test of the capability of agentic AI to augment and automate grant preparation”

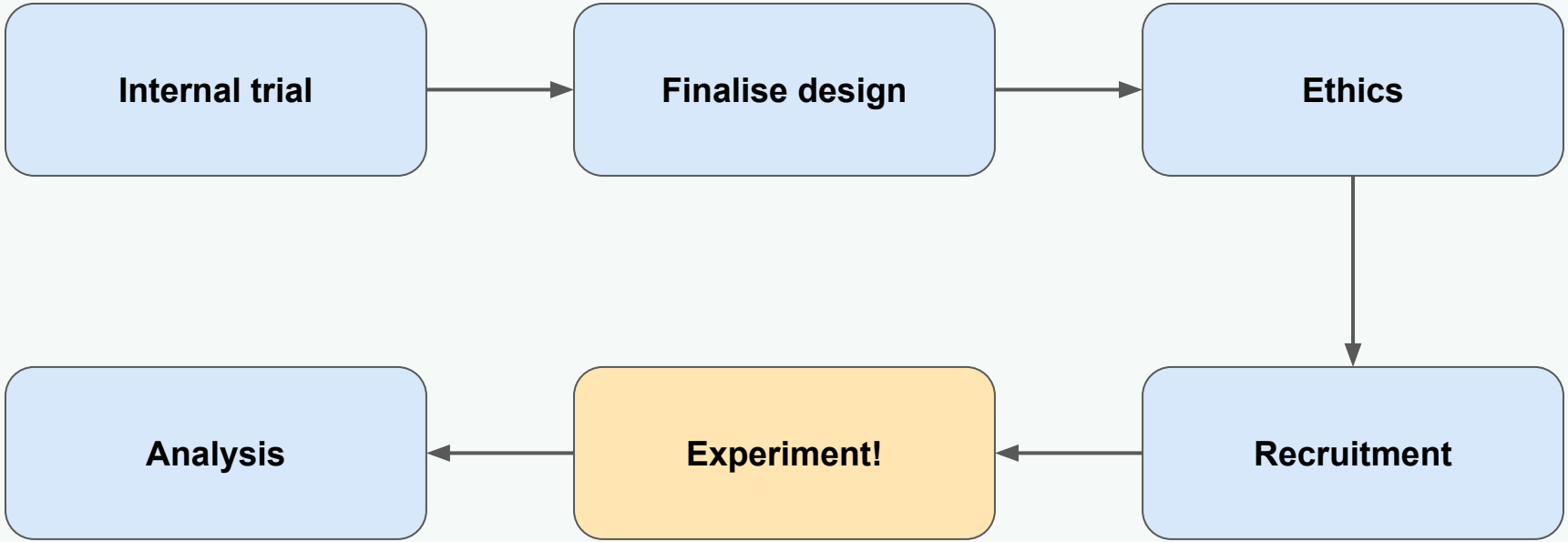
Specific researcher profiles

Specific funder and funding scheme(s)

Internal review + DPR by applicants (+ external reviewers?)

Full disclosure: applicants and staff (and external reviewers) must know that the proposals are i) part of an experiment ii) written by/with AI iii) not eligible for funding

Process


Internal trial
Finalise design
Ethics
Recruitment
Experiment!
Analysis

Open questions

Is full review (including external reviewers) necessary?

Disciplinary spread?

How to make sure we learn the most possible from the process?

What are outcome measures?

Can you participate?

Additional work? (UKRI: “a retrospective trial with the UK Metascience Unit to compare what AI would have concluded from past applications against the decisions that were made”)

Feedback: t.stafford@researchonresearch.org

The agentic turn in AI



For most of their short history, Large Language Models (LLMs) have been used interactively; a user poses a question and the model responds as a ‘chat bot’.

Agentic AI changes this relationship. An agent is an LLM equipped with tools such as the ability to search the web, read documents, write and execute code, or call external services.

Grant writing and reviewing sit naturally in the space of agents. The task is goal-oriented, information-rich, rule-bound and iterative. These are exactly the conditions where agents perform well.





ANTHROP\C

Agentic coding and persistent returns to expertise

Published
June 16, 2026

Authors
Zoe Hitzig, Maxim Massenkoff, Eva Lyubich, Ryan Heller, and Peter McCrory

https://www.anthropic.com/research/claude-code-expertise

Claude does more per prompt for more expert users

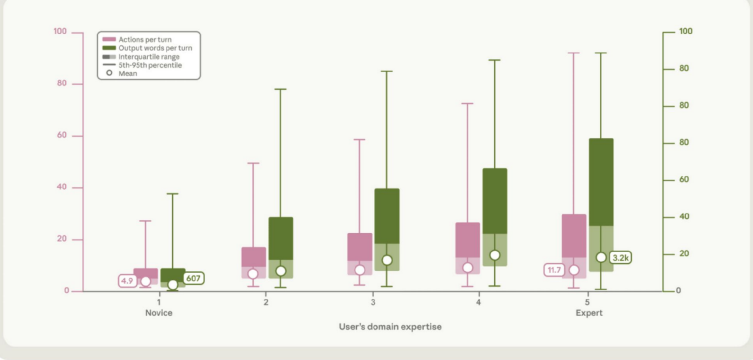


Figure 3: Claude does more per prompt for more expert users
Claude produces more actions (left bar) and text output per prompt (right bar) for more expert users. Boxes span the interquartile range (split at the median). Whiskers represent the 5th to 95th percentile. White dots are geometric means. Both upward trends are statistically significant ($p < 0.001$), as is each adjacent-level step, and they remain significant (at +9% actions and +13% output per expertise level) in a regression controlling for work mode, task value, month, occupation, and model family, with standard errors clustered by user.

September 6, 2026 Research Publication Safety

Research acceleration: The view inside OpenAI

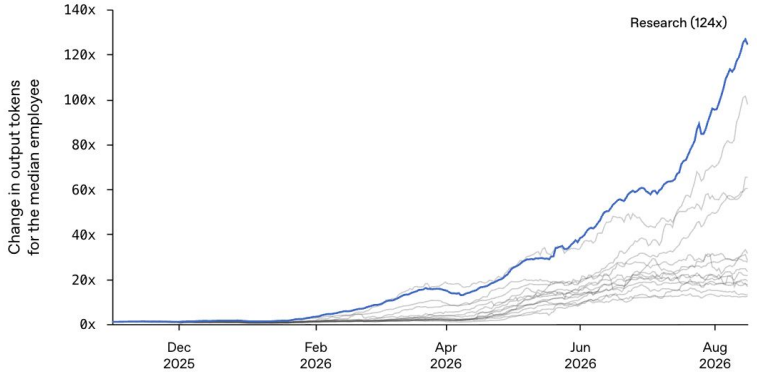
Listen to article 13:39 Share

For AGI to benefit all of humanity, we believe it must be democratically governed. This can only happen through an informed public debate about the capabilities, risks and safeguards of highly capable AI systems. People everywhere need to understand the likely future trajectory of frontier AI, so they can have a meaningful voice in how it develops.

Transparency about specific risks, incidents and safeguards is necessary, but not sufficient. We believe the public also needs to understand how the most capable systems are developing, and how they are driving research progress, inside of frontier labs.

We aim to safely build an automated AI researcher that can work under human supervision to further progress on deep learning and alignment, enabling iterative improvements. According to our measurements, we have now reached the goal, announced last fall, of having an automated research intern by September of this year. By

Usage growth is faster among researchers than in other parts of the company



View methods

Before June 2026, total agent runtime across the research organization was still below that of total human labor. That has since changed. In terms of a standard 8 hour workday, as of mid-August, in total, the research organization uses 3.1 agent-workdays of effort for every workday of human labor.

https://openai.com/index/research-acceleration-view-inside-openai/